ClaimGuard

Pay honest claims in hours. Stop fraud before payout.

Real-time AI risk scoring for insurance claims, inside the transaction on IBM Z

Live on IBM LinuxONE: change-illustrations-tooth-complete.trycloudflare.com

IBM Z Datathon 2026 | Theme 1: Real-time AI on Transactional Data

THE PROBLEM

Fraud and slow payouts hurt everyone

Priya: honest

Minor accident, Rs 18,000 repair.
Waits ~2 weeks while her claim
is checked by hand like everyone else's.

Rahul: suspicious

Rs 2.4 lakh 'total loss' on a 9-day-old
policy, no police report, same garage
as 7 linked customers.

Manual checks are slow and miss organised rings; red-flag rules raise false alarms.
Fraud losses become higher premiums, and honest customers pay twice: in money and in waiting.

OUR SOLUTION

Score every claim the instant it is filed

Fast-track payout

37% of claims

paid with zero manual work

Quick document check

59% of claims

one extra upload in the app

Send to investigator

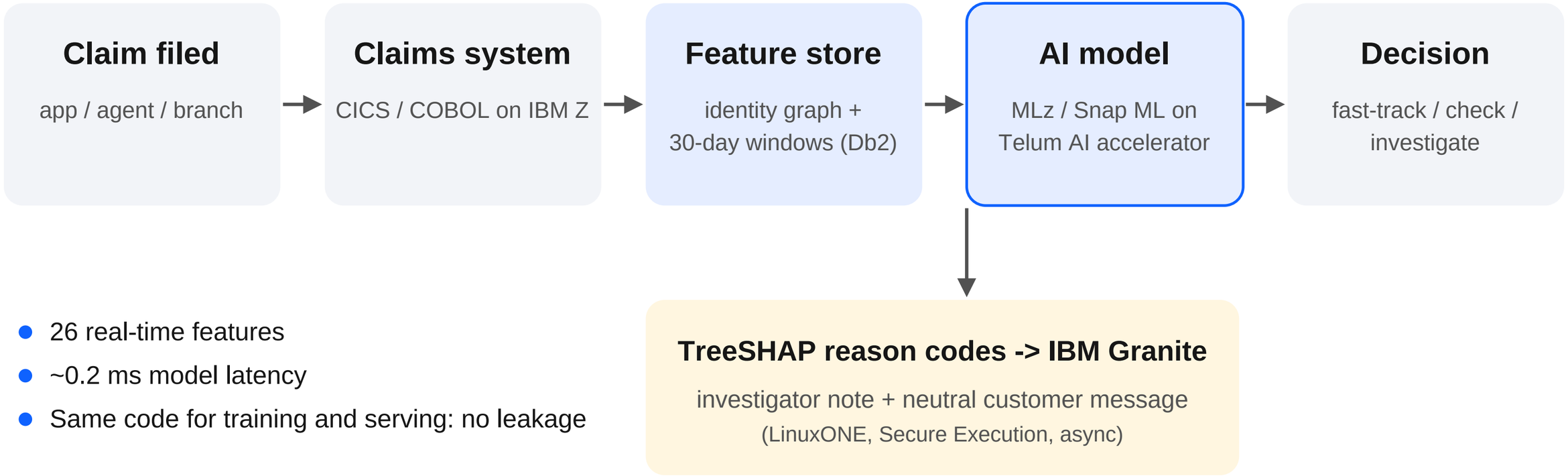
4% of claims

77% hit-rate, payout held

Every decision ships with exact reason codes (TreeSHAP) and an IBM Granite-written note.
No claim is ever rejected by a machine: humans decide every high-risk case.

ARCHITECTURE

How it works: AI inside the claims transaction



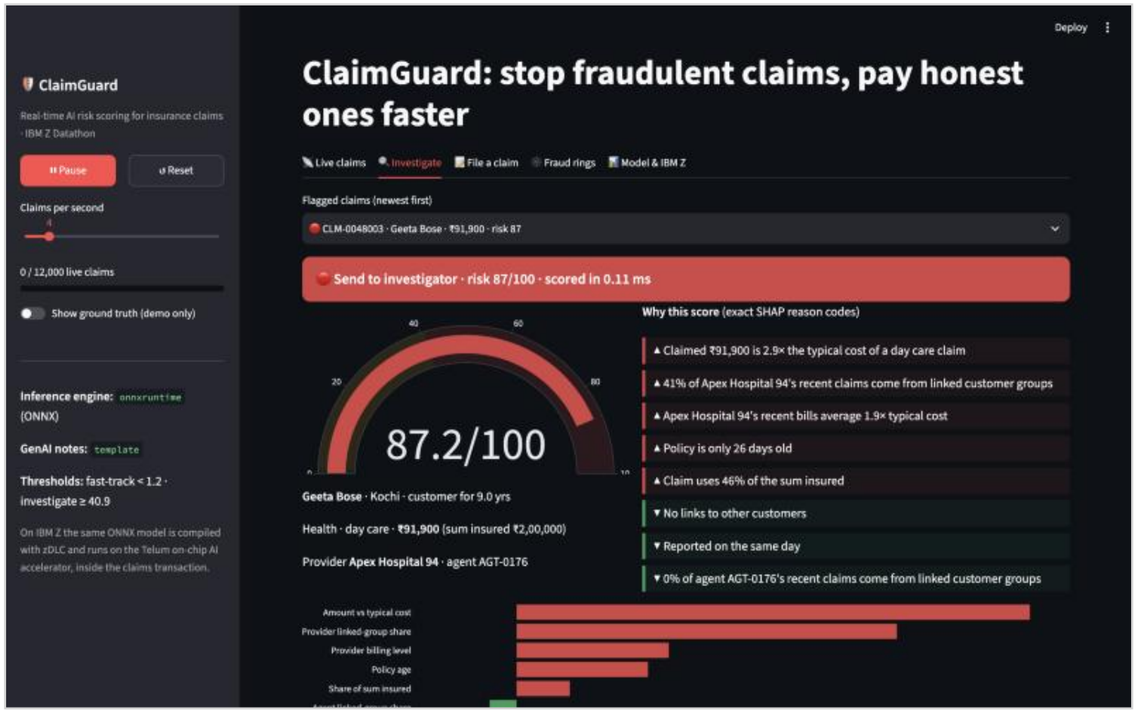
- 26 real-time features
- ~0.2 ms model latency
- Same code for training and serving: no leakage

PROTOTYPE

Live demo: scoring a stream of claims



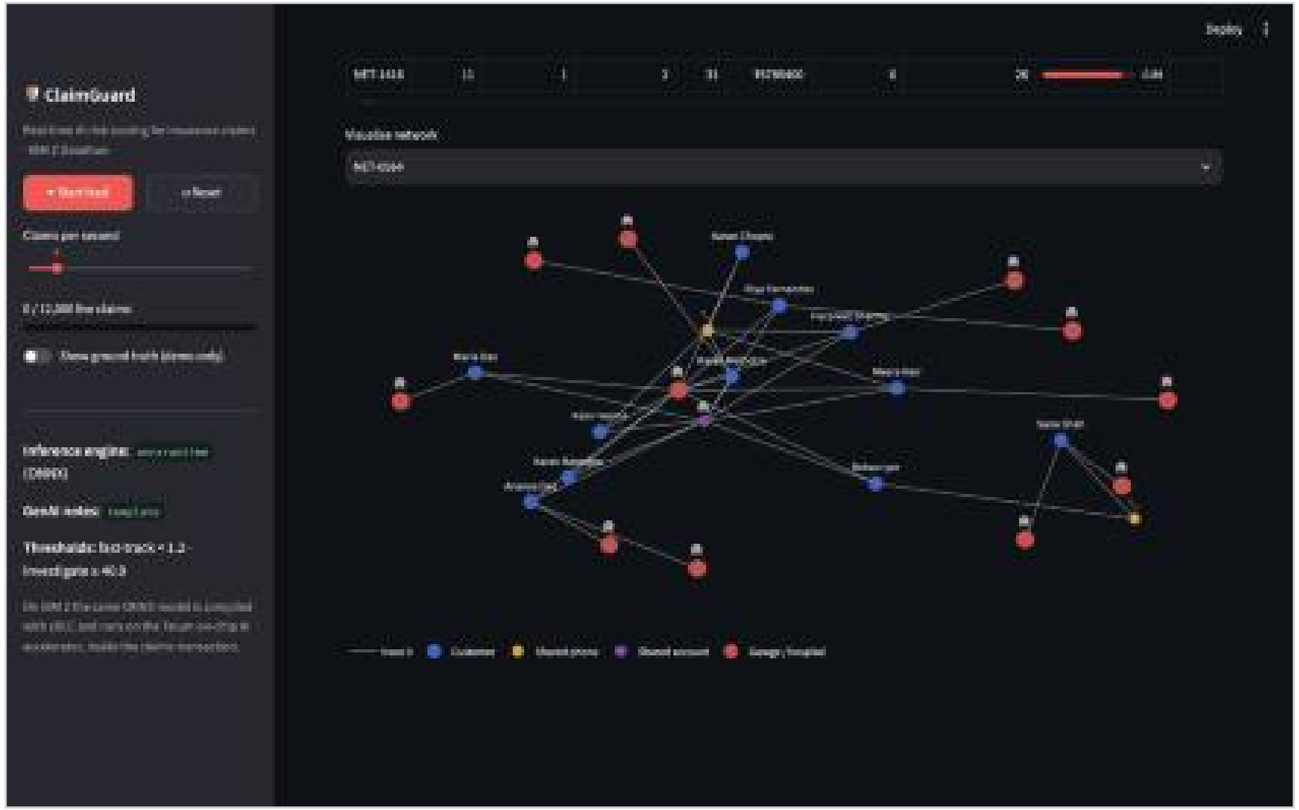
Live feed: every claim scored in ~0.2 ms, routed to a lane



Investigator view: exact reason codes for each flagged claim

GRAPH INTELLIGENCE

Seeing what rules cannot: fraud rings



- Linked by shared phones / accounts
- Claims funnelled to one garage
- Same agent, late-night incidents
- Graph updates in real time

Rings caught before payout:

100%

IMPACT, MEASURED

Results on the 12,000 most recent claims (time hold-out)

99%
fraud value stopped

37%
claims paid instantly

0.91
ROC-AUC

77%
investigator hit-rate

Same alert volume (2,336 alerts): share of fraud caught



PR-AUC 0.66 vs 0.07 random | only 3% of fraud reaches the fast lane | ONNX parity 3e-7

IBM Z INTEGRATION

Why IBM Z: intelligence where the transaction happens

Live on IBM LinuxONE (s390x)	trained + served on LinuxONE today, public demo
Telum on-chip AI accelerator	scores 100% of claims synchronously, no network hop
Machine Learning for z/OS	serves the model in-transaction to CICS/COBOL claims apps
IBM Snap ML	tree-model inference on Telum; imports our XGBoost model
Db2 for z/OS	feature store next to claim records, data never leaves
Pervasive encryption + Crypto Express	health & financial data encrypted, HSM keys
Secure Execution / Hyper Protect	Granite + dashboard in confidential containers

TRUST

Responsible by design, rigorous data

Responsible AI

- Human decides every high-risk claim
- Exact reason code for every score
- Neutral, non-accusatory customer messages
- GenAI explains; it never decides
- No protected attributes used

Data methodology

- 60k-claim simulator: 5 documented fraud schemes
- Overlapping distributions + 0.5% label noise
- Strict time split 70 / 10 / 20, no shuffle
- Thresholds set by business policy on validation
- Plug-in ready for real insurer data

VALUE

Impact and what's next

- Policyholders: paid in hours, not weeks
- Insurers: fraud stopped before payout, lower handling cost
- Investigators: short, explained queue with fraud-ring graphs
- Society: lower premiums, more trust, wider coverage

Try it now (running on IBM LinuxONE): change-illustrations-tooth-complete.trycloudflare.com

Next: real insurer data -> MLz / Snap ML in a CICS claims flow -> crop, life & travel claims

Honest claims paid in hours. Fraud stopped before payout.